

KUCHANA VARSHA

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Education

University of Missouri

Bachelor's in computer science

Columbia, MO

08/2024 - Present

SR International Institute of Tech

Bachelor's in computer science

INDIA

09/2020 - 05/2024

Experience

Undergraduate Research Assistant

Digital Biology Lab

Columbia, MO

2026 – 2026

- Analyze and process large-scale biological and sequencing datasets using Python and data analysis workflows.
- Support development and testing of FASTAptameR 3.0, a bioinformatics platform for high-throughput data analysis and visualization.
- Assist with data preprocessing, validation, clustering, motif analysis, and reporting workflows.
- Collaborate with researchers to improve data analysis performance, visualization accuracy, and user experience.

Peer Learning Assistant

University of Missouri, Department of Computer Science

Columbia, MO

2026 – Present

- Conduct office hours and provide technical support for students in HTML, CSS, and JavaScript.
- Assist with grading and student learning support via online platforms.

Projects

Retail Analytics Platform | Python, PostgreSQL, SQL, Tableau

- Designed and developed an end-to-end retail analytics platform by collecting product data from APIs and generating realistic sales, customer, and transaction datasets.
- Built ETL pipelines using Python, Pandas, and PostgreSQL to clean, transform, and load data for analytical reporting.
- Wrote advanced SQL queries to analyze revenue trends, customer behavior, product performance, and regional sales patterns.
- Developed an interactive Tableau dashboard with KPI tracking, sales forecasting insights, customer segmentation, and product performance analysis.

Driving School Management System | Python, MySQL, SQL

- Designed and managed relational MySQL databases for scheduling, billing, student records, and performance tracking.
- Developed SQL queries and reporting workflows to retrieve, analyze, and organize business data efficiently.
- Implemented data validation and data integrity checks to improve database accuracy and consistency.
- Automated data management processes using Python and SQL-based operations.

Parking Stall Occupancy Detection System | Python, OpenCV, NumPy

- Developed a computer vision and data analysis pipeline for parking occupancy detection using Python and OpenCV.
- Processed and analyzed image datasets using HSV features, edge detection, thresholding, and homography techniques.
- Generated structured CSV outputs for occupancy classification and reporting workflows.
- Improved data extraction and automation processes for large-scale image analysis.

iFINANCE Web Application | ASP.NET MVC, C#, SQL Server, Entity Framework

- Developed a financial management web application with SQL Server database integration and reporting functionality.
- Designed normalized relational database schemas and implemented secure role-based access controls.
- Built data-driven reporting features for financial analysis and transaction management.

LaserPurrssue: AI Powered Smart Laser Control System | Kotlin, Jetpack Compose, WebSockets, Media3, Raspberry Pi

- Developed the Android frontend for an AI powered smart laser system with real-time remote monitoring and control.
- Built low-latency device communication using WebSockets, Jetpack Compose, and RTSP video streaming with Media3.
- Collaborated on Raspberry Pi backend integration using Flask/FastAPI and OpenCV-based cat detection workflows.

Technical Skills

Data & Analytics: Power BI, Tableau, Oracle Analytics Cloud, Microsoft Excel, SQL, Data Visualization, Reporting, ETL Concepts

Languages: JavaScript, TypeScript, SQL, Python, Kotlin, C, C++

Web & Mobile: Angular, React, ASP.NET MVC, Node.js, Express.js, Jetpack Compose, HTML, CSS, REST APIs, WebSockets

Databases: PostgreSQL, SQL Server, MySQL, MongoDB

Tools & Frameworks: Flask, FastAPI, Pandas, OpenCV, Media3/ExoPlayer, TensorFlow Lite, Git, Docker, VS Code, Visual Studio